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Daniel A. Delaurentis & Donald N. Fry

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ARTICLE

Understanding the Implications for Airports of Distributed Air Transportation Using a System-of-Systems Approach

DANIEL A. DELAURENTIS & DONALD N. FRY

School of Aeronautics & Astronautics, Purdue University, West Lafayette, IN, USA

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ABSTRACT This paper merges two perspectives to better understand the future requirements for airports. From the methodological perspective, a system-of-systems (SoS) approach is employed to provide structure to the numerous, multi-level aspects crucial to understanding air transportation dynamics. From the operational perspective, distributed on-demand air service is introduced as a new service paradigm. This new service intends to improve doorstep–destination mobility for regional intercity trips. Its realization could dramatically shift the loading on the present day network of airports. To examine the potential impacts, network models are developed from the SoS foundation for the new on-demand service, a hub-and-spoke network topology, and a scheduled point-to-point service. An example is provided which illustrates the implications of these service models on airport fitness using Indiana airports as a case study. The ability of an airport to be part of many preferred trips, its network centrality, is found to drive fitness in distributed service models. The overarching objective of the approach is to provide decision support for airports that integrate all relevant dimensions – economics, technology, operations, and policies.

KEY WORDS: Airports; requirements; on-demand service; system-of-systems; networks

Correspondence Address: Daniel A. Delaurentis, School of Aeronautics & Astronautics, Purdue University, West Lafayette, IN 47907, USA. Email: ddelaure@purdue.edu

Introduction & Motivation

The purpose of an airport is to act as a portal that enables the transport of people and goods by air. In fulfilling this purpose, the hallmark of a successful airport can be encapsulated in four pillars put forth by Louis Turpen (who has been chief executive of both San Francisco and Toronto airports over the last 10 years) (Turpen, 2004): *safety & security*, *customer service*, *environmental sensitivity*, and *financial independence*. While these pillars are generally relevant and applicable for all airports regardless of size or nature, particular requirements in each of the four may vary significantly for an individual airport. The nature of financial independence, for example, is different in a private versus public model. Furthermore, these requirements may shift significantly over time due to changes in the nature, number, and intent of entities with which an airport interacts. One of the primary sources of such a shift is the introduction of new air transportation service concepts. Determining the best set of requirements in this context remains a challenge.

Ideas for new air transportation service concepts are constantly emerging. These ideas are motivated by a desire to provide more efficient and attractive doorstep to destination travel, and a need to relieve over-stressed hub airports which will face even greater challenges due to the projected increase in demand for air transport (a 40% increase from 2005 to 2017 according to a recent FAA forecast (FAA, 2006)). Stress on the system, of course, translates to passengers, with increasing frustrations that are becoming common stories in the popular media (Levin, 2007). One new paradigm on the horizon, in particular, emphasizes a distributed approach, seeking to satisfy travel demand across more airports via point-to-point routings. Furthermore, this concept pursues a flexible service offering that is available according to the needs of travelers, unlike traditional operations which are scheduled far in advance. This paradigm shift represents a transformation in operations that will force airports to respond in order to prosper and participate in these new services.

Three specific concepts are under development, or have recently been completed, which may contribute toward creating a distributed, on-demand air transportation. The first is the NASA Small Aircraft Transportation System (SATS) project, which has recently concluded. This project sought to develop the concept of operations and technologies that would allow a dramatic increase in utilization of currently under-utilized small- and medium-sized airports (Holmes *et al.*, 2004; Lohr *et al.*, 2005). SATS research demonstrated technologies involving both airports and aircraft avionics that could potentially increase the amount of reliable operations at such airports.

Motivated by the feasibility demonstrated by SATS, a second concept is attracting a growing number of potential operators who are planning to offer on-demand air service (ODAS), often loosely called ‘air taxi’. ODAS is a business model that would make use of the SATS infrastructure to provide on-demand, point-to-point air service on a regional basis. The third concept is widespread use of personal air vehicles (PAV). The PAV concept looks beyond ODAS to the notion of large-scale use of personally piloted aircraft for truly on-demand, reliable, doorstep to destination air transportation (Moore, 2006).

Either ODAS or PAV, and certainly both together, would represent a major paradigm shift for air transportation. They seek to provide a new mobility resource that is superior to the scheduled, commercial aviation, and auto options for many trips. Figure 1 shows three different options for traveling from origin to destination. The Airline and Ground Transport modes require significant indirect travel and low speeds with multiple refueling stops, respectively. An alternate mode, such as ODAS or PAV, may include much shorter ground travel to and from air portals and more direct flight segments by utilizing smaller, regional airports. Increasing the number of flights to smaller airports in order to reduce the airborne portion of travel time, however, is only one aspect for reducing doorstep-to-destination (DD) time. Regional planning to coordinate *multimodal aspects* will also be a key aspect for realizing improvements in the DD trip. Air-to-ground passenger interchange and corresponding ground transportation must be accounted for when accessing these smaller airports, especially since they may have reduced access to mass transit, rail, or taxi/rental car services compared to large, urban airports. Overall, the effectiveness of

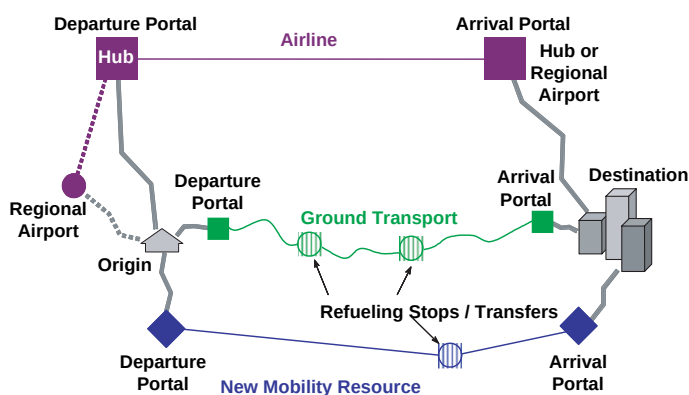


Figure 1. Doorstep-to-destination motivation for new distributed air transportation concepts (Lewe *et al.*, 2006)

an airport in the midst of ODAS and/or PAV is a function of how well it enables many travelers to complete their preferred DD trip.

The viability of these distributed air transportation concepts rests on three developments: new aircraft technology, new business models, and flexible operational and management policies (which permit optimal use of existing and new infrastructure resources). The first development is already quite far along, with a new generation of small, affordable jet aircraft termed 'very light jets' (VLJs) having been designed, marketed, and certified. These aircraft generally weigh less than 4500 kg at takeoff, seat 4–6 passengers, provide significant speed advantage (upwards of 40%) over similar sized turboprop aircraft, and require much shorter runways than larger jets. Consequently, VLJs can use the more than 3400 local/satellite US airports identified under the SATS project, substantially increasing geographical accessibility. Several VLJ models are already in production, notably the Eclipse Aviation Eclipse 500 (Harness, 2003).

New business models have also been proposed that seek to exploit both the new aircraft technology and the market needs for more attractive solutions. The price point in these new models is below those of the corporate jet or fractional ownership markets. Indeed, some are characterizing their operations as 'air-taxi' and examples exist of regional services, which intend to employ small, un-crowded airports close to where people live and/or work. For example, the DayJet Corporation plans to inaugurate service this year to five Florida cities utilizing the Eclipse 500 jet (Sharkey, 2006). It is quite plausible that as such service is proven to be safe, effective, and profitable, the ODAS will improve and expand to include a wide range of citizens who can afford the utility that the service brings.

The status of the third required ingredient for successful growth – proper operational and management policies – remains less clear. What is clear is that the successful fruition of distributed air transportation potential requires a holistic treatment of relevant effects and dynamics. Adequate and convenient interconnection with other modes, proper access within communities of users, and adequate attention to environmental considerations are key issues. In addition, the strategic understanding of its role and potential evolution relative to the National Airspace System (NAS) is also crucial. Effective policy decision-making requires that stakeholder groups have an informed understanding of these dependencies.

In this paper, we present a system-of-systems (SoS) approach that is intended first and foremost to provide structure to the range of problems involved in analyzing future requirements for airports. This type of approach considers the relevant adjacent systems and sources of uncertainty associated with the new, distributed-type air service. In

particular, well-defined analysis problems can be posed that consider an airport as a node in a set of interrelated networks that evolve over time. Network topology analysis is suggested in order to determine airport fitness among communities of other airports and in the presence of new/emerging operational concepts. Simulation results that utilize network models in the presence of evolutionary mechanisms that mimic ODAS and PAV behaviors within Indiana are reported to further illuminate the methodological ideas. The ultimate objective is robust decision-making to determine technological, operational, or policy-related changes airports must make in order to thrive in a more distributed-load operating environment.

Problem Background

Data and Operational Aspects

Depictions of representative vehicles in both the PAV and VLJ categories are shown in Figure 2. NASA's Civetta concept vehicle employs a tailfan propulsion system for noise management and emphasizes novel manufacturer constructs to achieve a highly affordable design. A broader class of PAV vehicles has also been considered. These PAV options have been categorized into four groups based on their takeoff and landing field length distances: vertical takeoff and landing, VTOL (30 m), extremely short takeoff and landing, ESTOL (150 m), short takeoff and landing, STOL (300 m), and conventional takeoff and landing, CTOL (900 m). Field length is the key linkage between the aircraft and the airport, and often dictates the mission a given aircraft is capable of performing. Vehicles with lower field length requirements may allow for air portals to be utilized which are much closer to where people live and work than larger airports. Furthermore, results from a recent stated-preference air transportation survey found

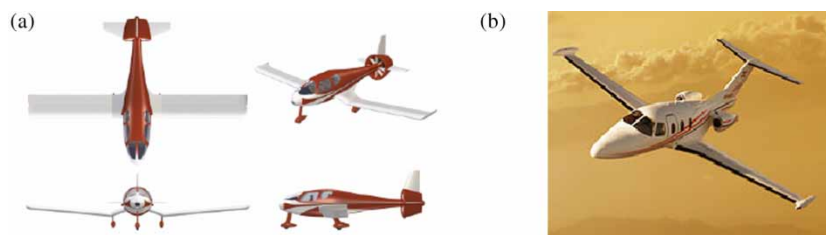


Figure 2. Two aircraft proposed for use in distributed air transportation: (a) NASA's EQuIPT Civetta PAV demonstrator concept (Hahn, 2006) and (b) prominent VLJ: Eclipse Aviation Eclipse 500 Jet (Eclipse Aviation, 2007)

that, along with fare and distance, workplace accessibility of ODAS was a significant factor in determining whether travelers would be willing to switch to the new service from their current preferred mode (Peeta *et al.*, 2007). While ODAS proximity to work is an attractive attribute for users, environmental concerns can arise if it is close to residential areas. In this context, the acceptability of VLJ noise levels must be (but has not yet been) determined, providing further uncertainty in the design of future VLJs.

Runway layout is also an important consideration in revolutionizing future airport architectures, and optimal requirements may vary with regard to VLJs and PAVs. One strip aligned with the prevailing winds is sufficient for airfields that are primarily recreational in nature. More reliable availability likely requires at least two strips at different angles (to cover a broader range of wind conditions). If absolute reliability is required, then the classic Navy three strips arranged in a triangle are needed. An additional obstacle to increased utilization of small airports is that many lack the infrastructure to allow precision instrument approaches. Research under SATS sought means to reduce this barrier, and Xu *et al.* (2005) found that increased use of Wide Area Augmentation System (WAAS) could produce some of the desired improvements. However, they also identified a need for cost-to-benefit studies of the various options, indicating again the competing effects brought to bear when exploring airport requirements for distributed air transportation.

Challenges will also appear in the airspace proximal to airports. Increased activity around under-utilized airports will increase the complexity of operations, likely requiring greater pilot-to-pilot coordination. One aspect of the SATS program was designed to address this very issue in order to coordinate numerous landing operations between pilots in the airspace around an airport, without increasing the workload of the pilots (Lohr *et al.*, 2005; Consiglio *et al.*, 2005). Furthermore, a recent study indicates that, as core regional airports saturate, reliever airports will become critical for accommodating traffic demand. If VLJs are pursued as a solution, this may significantly increase the traffic load at the terminal airspace and these impacts need to be taken into account in transforming the NAS (Bonnefoy & Hansman, 2007).

Adjacent Systems

The operational aspects summarized in the last section highlight many of the systems 'adjacent' to the airport and suggest a tight coupling that exists among them. Several of these adjacent systems, and their influence in the context of distributed air transportation, are shown

Table 1. The influence of systems ‘adjacent’ to airports

Adjacent system	Influence on airports in the context of distributed, on-demand air transportation
Terminal Area Airspace System	Limitations on high volume operations at smaller airports
Aircraft Systems	Operational needs of new aircraft (e.g. VLJs)
Ground Transport Systems	Enables doorstep-to-destination connectivity
Air Service Providers	Needs of customer stakeholders
Other Airports	Enables regional airport service clusters
National Airspace System	Requires compliance with nationwide operational rules
Local/Regional Government	Partnering to address environmental concerns
Local Economy	Economic drivers (e.g. regional revitalization is a key motivation for SATS/ODAS)

in Table 1. These adjacent systems along with the airport form a community of interacting systems. Effective development of requirements for airports requires a holistic treatment of the influences between these adjacent systems. By characterizing the relationships represented in the various networks, a holistic understanding of the endogenous and exogenous entities can be formed (Lewe *et al.*, 2004).

Need for a New View

Cognizant of the adjacent systems, an approach is desired with integrated measures that captures the fitness of an airport and thus serves as a decision-making tool for airport planning. The emerging field of SoS offers a source for hypothesized approaches. An SoS consists of multiple, heterogeneous, distributed systems embedded in networks at multiple levels that evolve over time. Thus, SoS methodologies focus upon the dependencies and interconnectivities between systems which collaborate, but also which may operate autonomously. By expanding the analysis boundary to include both technical and non-technical facets, a more complete understanding of the system can be developed. However, this pursuit likely brings additional complexity. Complexity in an SoS stems primarily from the heterogeneity of its constituent systems, the distributed nature and organization of these systems, and the presence of deep uncertainty with respect to its future state.

The structure of organization is key to managing complexity in SoS problems. Herbert Simon (1996), Nobel Laureate and artificial intelligence pioneer, asserts that ‘resemblance in behavior of systems without identity of the inner systems is particularly feasible if the

aspects in which we are interested arise out of the *organization* of the parts, independently of all but a few properties of the individual components.' A SoS approach seeks to structure organization in a manner that exposes the multiple levels of abstraction that are present. In particular, the structure represents connectivity of the multiple, heterogeneous, participating systems in the context of stakeholder and exogenous influences and distinguishes holistic analysis from more traditional system analysis.

In considering future roles of airports, segregating the analyses to the individual systems which interact with the airport removes important information from the SoS organization. In particular, the inappropriateness of segregated airside and landside analysis is accentuated in new concepts, such as ODAS and PAV, which rely on multimodal capabilities. While both seek to use the air mode to get passengers closer to destinations, they still rely on the ground modes (which may not be in place at smaller airports). The following section will establish a framework for analyzing airports in the face of a changing operational environment from a variety of contexts.

System-of-Systems (SoS) Approach

Articulating Problem Structure

In order to integrate and organize the systems from Table 1, a flexible lexicon has been formed (DeLaurentis & Calloway, 2004). Because systems can be aggregated in a hierarchical manner, the lexicon enumerates these interacting *levels* of various components with Greek letters. Starting from α , the levels proceed upward, leading to γ or δ or higher, depending on the complexity of the problem and the depth of analysis desired. Furthermore, a β -level constituent represents a network of α -level entities; a γ -level constituent represents a network of β -level entities, and so on.

In addition to these hierarchical labels representing levels, a set of dimensions are defined that highlights the trans-domain aspects of SoS. Specifically, not all entities within the levels are similar in their basic character; they can be made of entities or systems with different characteristics, representations, and from different domains. Thus, a framework is needed to classify the different components of each level. The scope of system domains for air transportation can be categorized primarily into *Resources*, *Operations*, *Economics*, and *Policy*. Each of these dimensions separately comprises the previously described levels, thereby completing the SoS lexicon. The categorization of the levels lends clarity and organization in dealing with the different facets of the problem while maintaining the lucidity provided by the levels. The

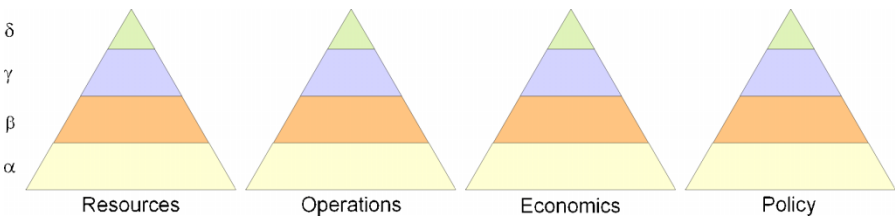


Figure 3. Scope dimensions and hierarchy of levels: an unfolded pyramid unifying the lexicon

relationship between dimensions and levels is shown in Figure 3 and described in Table 2.

Employing this lexicon, the airport can be couched in terms of its location in SoS space, as illustrated in Table 3, which presents an expansion of the lexicon for the transportation system. Once formed, this expansion exposes the context-specific relationships in which the airport is involved. Specifically, an airport is a β -level construct in our SoS lexicon. It, itself, is a SoS as well as being a larger part of the SoS around it. As a β -level entity, the airport (and any useful analysis of it) has to take into account γ , δ , and α levels and its influence across all of those levels. For instance, α -level resources such as aircraft, ground service vehicles, gates, and runways comprise the physical manifestation of an airport. The common functioning of these systems is shaped not only by the desires of the owners of this network but also the interoperability rules that arise from the operations, economics, and policy scope dimensions.

While physically adjacent systems are certainly important to the operations of the airport, the concept of adjacency takes on a new meaning in terms of connectivity across dimensions and between levels. Because the airport does not operate in isolation, a mapping of the connections allows the relevant networks to be specified. Subsequently,

Table 2. Description of SoS lexicon categories

Dimension	Description
Resources	The entities that give physical manifestation to the system-of-systems
Operations	The application of policies/procedures to direct the activity of physical entities
Economics	The sentient systems (individuals and organizations) that give a ‘living system’ character to the operation of the physical entities
Policies	The set of rules that bound functioning of entities in all dimensions

Table 3. Use of system-of-systems lexicon for air transportation

		System-of-systems dimensions			
	Level	Resources	Operations	Economics	Policy
Base level	α	Vehicles & infrastructure (e.g. aircraft, runway, terminal area)	Operating a resource (e.g. aircraft, control tower)	Economics of building/operating/buying/selling/leasing a single resource	Policies relating to single-resource use (e.g. flight procedures)
Network of networks	β	Collection of resources for a common function(e.g. airport)	Operating resource networks for common function (e.g. airline)	Economics of operating/buying/selling/leasing resource networks	Policies relating to multiple vehicle use (e.g. local airport noise policies)
	γ	Resources in a transport sector (e.g. air transportation system)	Operating collection of resource networks (e.g. commercial air operations)	Economics of a business sector (e.g. airline industry)	Policies relating to sectors using multiple vehicles (e.g. FAA certification)
	δ	National Transportation System	Operators of Total National Transportation System	Overall Transportation Forecasts/Market	National Transportation Policies
	ε	Global Transportation System	Global Operators in the World Transportation System	WTO, Global Marketplace	Global Transportation System Policies

the networks and their context can be used to separate, formulate and solve particular problems. Finally, we note that the four scope dimensions are closely aligned with the four tenets which airports must address for success: safety & security (operations), customer service (resources), environmental sensitivity (policy), and financial independence (economics).

Problem Analysis Approaches

Exposition of the airport within the multi-layer networks that comprise the SoS indicates that traditional systems analysis techniques must be reexamined for adequacy. First, there are two fundamental approaches for addressing multi-layered aspects: a bottom-up and top-down tact. Second, there are two primary avenues for dealing with interconnectivity: a treatment of a particular entity in isolation with any interconnectivity represented as an externality or treatment of the entity in the context of its network (or networks). Our choices in these two matters are briefly described next.

Top-down Approach vs. Bottom-up Approach

Systems engineering frequently tends to favor a bottom-up approach to analysis. Individual systems analyses can be used to investigate properties of and interactions between α -level entities. These properties and interactions are then often used to build metamodels which are used in β -level analyses. One perceived advantage is that this approach can employ high fidelity modeling and analysis programs; however, some of the value in this fidelity is lost in building the metamodels which are approximate by nature. In addition, metamodels may not be extendable to higher levels because of a mismatch with modeling assumptions implicit in higher level models. This approach might also be categorized as a 'segmented' approach, which can lead to inappropriate results or findings for a SoS problem.

In contrast to the bottom-up approach, working from a high(er) level, in particular with a holistic perspective, has several important features and capabilities. First, it is more flexible in that it does not require complicated α -level analysis packages. Furthermore, it is an extensible framework and hence, more germane to the organization of upper levels. In addition, it is a cross-disciplinary methodology that allows disparate domains (e.g. IT, human factors, policy management) to be considered and incorporated into the analysis. It is this type of top-down approach which we adopt in our investigation of the evolution of requirements for future airports.

Network Perspective vs. Isolated Analysis

In light of the heterogeneity of adjacent systems (Table 1) and the scope indicated in Table 3, we conclude that the number and nature of interactions that airports have with other entities demands direct treatment via a network perspective. This construct is developed further as shown in Figure 4 which displays the airport as a node in several networks. Each network is labeled with the scope dimension under which it primarily falls (although network constituents can span all four dimensions). For example, in a commercial economic network, an airport is a source of revenue and employment as well as a hub through which products travel (note the multidimensionality of the airport node even in numerous economic networks). This latter aspect is tied to the airport’s involvement in the ground transportation (resource) network in which goods and people travel to and from the airport node via the ground transport network. In a certain sense, the airport is both a destination node – one leaves ground transport at the airport – and an origination node – one reenters the ground transportation network after completing air travel. Similar delineations can be shown for the airport’s presence in

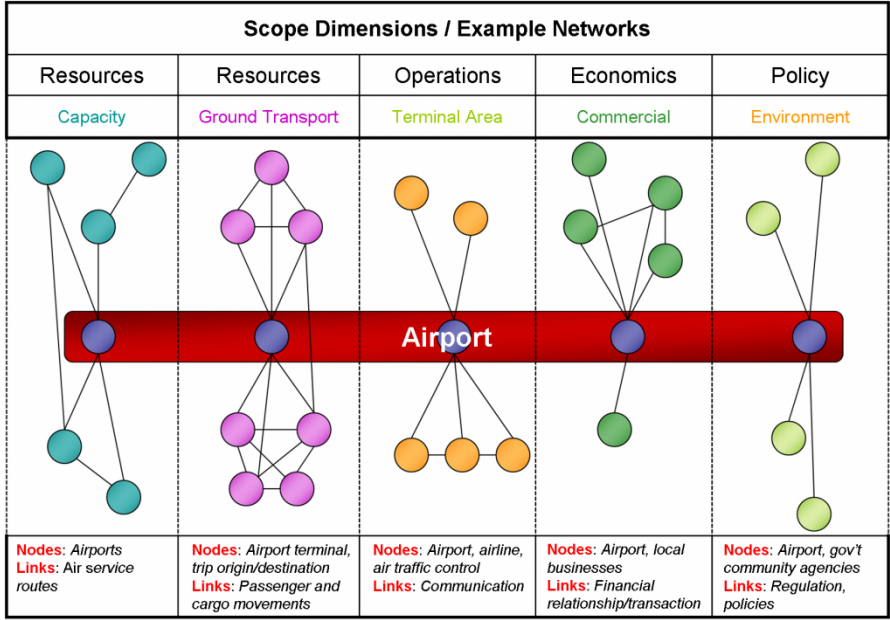


Figure 4. A γ -level view of an airport as a member of several β -level networks

policy and operational networks, as well as other resource and economic networks. Realistically representing and analyzing interactions within and among the disparate networks is the backbone of SoS analysis. It is hypothesized that such analysis can generate new important insights, for improvement in complex endeavors such as air transportation.

Airport Fitness in a Network Context

Establishment of relevant networks is only the first step in the SoS analytical process; models that explore the evolution of these networks under a variety of scenarios are also necessary. A key to evolution is the traits of the nodes, in other words, their *fitness* as members of the network. In our particular application setting, a fitness measure for airports is established. Because there are many stakeholders in any given SoS, the fitness measure is context dependent. For example, in our previous work, a nodal saturation measure was established which evaluated the ability of an infrastructure provider to maintain adequate capacity at an airport (DeLaurentis *et al.*, 2006). Additionally, airport fitness may depend on its relation to other nodes, e.g. the ‘community of airports’ which a particular airport connects to via service provider routes. Such measures and relationships define performance on the various networks of which the airport is a participant.

The field of network science has produced a number of generic measures for evaluating nodal importance in networks (Newman, 2003). Measures of nodal importance include degree (the number of links a node has), the strength (number of links combined with the weight of the links), the clustering coefficient (a measure of the cohesiveness of a set of nodes), and the centrality (the relative importance of a node in terms of connecting otherwise disconnected nodes). Several different centrality measures have been defined in the literature. One of these measures, betweenness centrality (a measure of the number of shortest paths of which a node is a member), has been used to characterize the importance of certain nodes (airports) in bridging communities within the US air transportation network (Han & DeLaurentis; Guimerà *et al.*, 2005). Another is the eigenvector centrality, which assesses a node’s strength within a community of surrounding nodes. Generally speaking, if an airport is ‘highly central’ in the capacity network, for example, this indicates that the airport is integral for connecting otherwise separated sections of a network. We define in more detail and explore several of these measures in the next section.

Case Study: Indiana Airports

Problem Definition

We present a case study to demonstrate how a particular exploratory analysis can be framed and conducted via the SoS approach. This case study focuses on an assessment of airports in a statewide region in the midst of distributed air service and addresses interactions of β and γ -level entities. Since distributed, on-demand air transportation is largely a new type of service for which little data exists, we utilize a simulation-based approach in which the capacity network (left most network described in Figure 4) is evolved under differing evolutionary mechanisms that mimic operations, such as ODAS and PAV. Additionally, the ground transportation network interactions are represented in a simple way by a ground efficiency parameter.

Air transportation service networks can be characterized along two key dimensions: topology and schedule. The network structure of mainline carriers in the current capacity network in the US is generally considered as hub-and-spoke, and thus is topographically similar to what network science calls a scale-free network (Barabási & Bonabeau, 2003). In this type of network, nodes and links are added following a ‘preferential attachment’ scheme. This means that when a new node is added to a network, it is more likely to add a link to a node which already has a higher number of links, compared to one with relatively few links (a kind of ‘rich get richer’ mechanism). In a point-to-point network, links are created on a more random basis; thus, most nodes will have approximately the same number of links. Although the two network types differ in topology, they share the trait that their schedule of service makes the topology static. By relaxing the fixed scheduling constraints on a point-to-point topology, one obtains a mechanism similar to an on-demand-type service. The key difference between a point-to-point scheduled network and an on-demand network is the time scale along which the network exists. An on-demand carrier network will be very dynamic, changing its schedule from day to day, since a daily network is merely the aggregation of all flights on that day. An aggregate PAV SP would be a similar network, but perhaps with a larger number of operations. Research funded by NASA has already demonstrated some gross estimates of the number of PAV operations possible in a regional airspace (RhinoCorps Ltd., 2006). A similar analysis of the impact on the European Air Traffic Management system has also been performed (Rohacs *et al.*, 2005).

The region chosen for this exploratory experiment is the state of Indiana, which has over 60 airports (see Figure 5(a)) and 6.1 million residents (US Census Bureau, 2000) (distribution shown in Figure 5(b)).

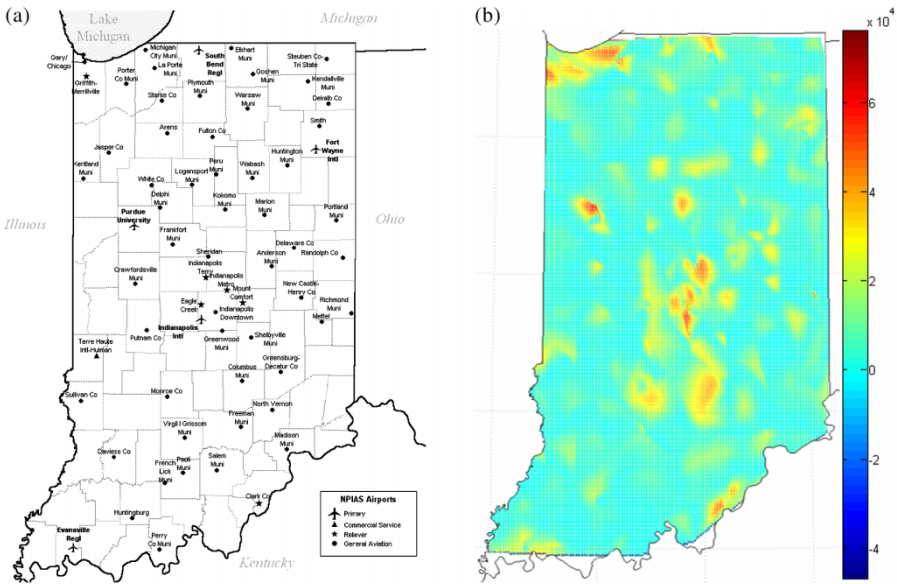


Figure 5. Indiana airports and population distribution: (a) location and type of Indiana airports and (b) population density of Indiana

It is quite easy, of course, to later include airports from adjacent states. However, such inclusion may not be so easy in *actuality* since airport planning and improvement programs are generally done separately by individual states. This is yet another problem within our SoS that could be framed and addressed in a second study, emphasizing relationships among regional planning policy aspects.

Models for Network Growth and Nodal Fitness

To investigate the impact of these three types of network evolution mechanisms (hub-and-spoke, point-to-point, and on-demand), the simulation employs the airport fitness measures introduced in section ‘Introduction & Motivation’. The growth mechanism for hub-and-spoke networks is related only to the number of links a node (airport) has. Hence, evolving a hub-and-spoke network is quite straightforward. First, a simple random network is formed with only a few links. Next, nodes are added and form links to existing nodes probabilistically, based on the degree (number of links) the existing node has. A probability distribution is calculated for each node (l) in the network based on the ratio of its degree (k) to the total number of links in the network, as defined below:

$$p_l = \frac{k}{k_{\text{total}}} \quad (1)$$

If p_l is less than a uniformly random sampled number between zero and one, a link is formed. Using this simple growth and attachment model, a hub-and-spoke network can be created. Again, in this case, the measure of nodal fitness is based only on the airport's degree. The degree distribution for this hub-and-spoke network is shown in Figure 6(a), while a visualization of the topology is shown in Figure 7(a).

Forming a network representative of a point-to-point service requires formation of an appropriate nodal fitness function. In an effort to more fully capture the communities of which the airport is a part, five parameters will be used to characterize the airport's fitness (see Table 4). Three of these parameters measure network properties and rely on the adjacency matrix A , which defines the connectivity between nodes. The adjacency matrix is an $n \times n$ matrix where element (i,j) is one (or any other positive number) if there exists a link between node i and node j , and zero if there is no link.

These values are normalized to allow for simple summation in a generic fitness function (η). This function is defined in Eq. (2) (shown in Figure 8). Note that the ground transport efficiency for this experiment is a random number scaled to the same relative magnitude of the other

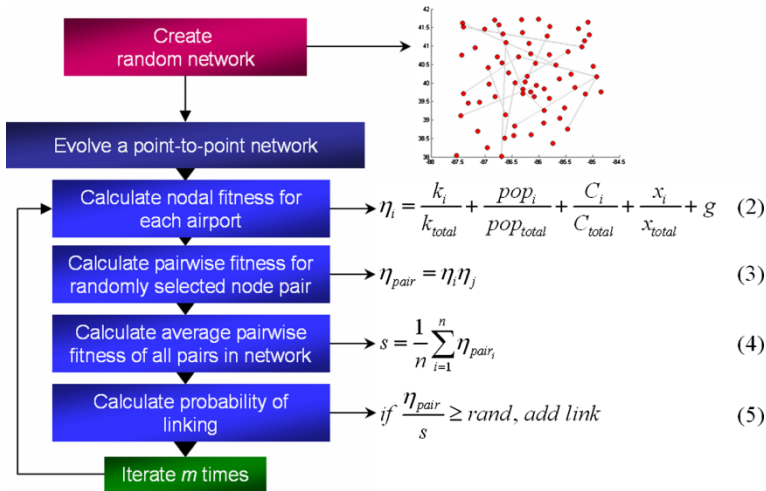


Figure 6. Algorithm (with fitness function definition) for evolution of point-to-point service networks

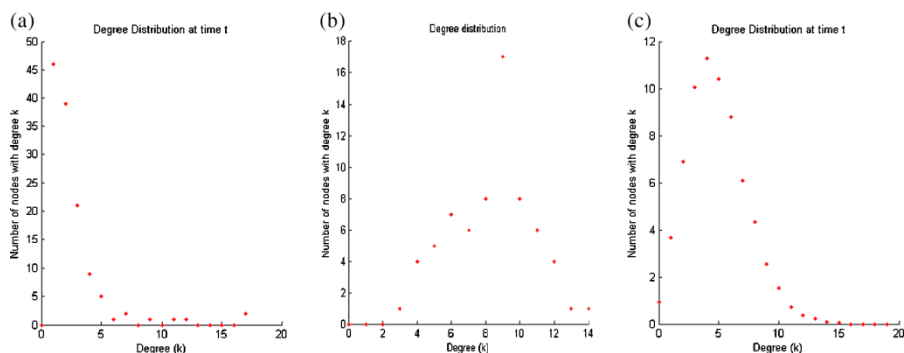


Figure 7. Degree distributions for the three types of networks: (a) hub-and-spoke; (b) point-to-point; and (c) ODAS

four parameters. Further studies can ascribe actual values for this parameter-based data from the airports in question.

The algorithm for network evolution for point-to-point service using the airport fitness function is shown in Figure 8. The process begins with creation of a random network of airports with very few links. Just as in the hub-and-spoke network, the point-to-point service network is

Table 4. Nodal characteristic measures for fitness computation in a point-to-point service network

Parameter	Symbol	Significance/Relevance	Equation
Degree	k	Surrogate for ‘popularity’ of location	$k_i = \sum_j A_{ij}$
Population	pop	Surrogate for customer demand at node	—
Clustering coefficient	C	Represents the connectedness of the cluster of which the node is a part	$C_i = \frac{1}{k_i(k_i - 1) \sum_{j,k} A_{ij} A_{ik} A_{jk}}$
Eigenvector centrality	x	Represents the strength of connectivity of node AND the nodes to which it links directly	$x_i = \lambda^{-1} \sum_j A_{ij}^w x_j$
Ground transport efficiency	gnd	Represents the efficiency of the ground transportation to/from node	—

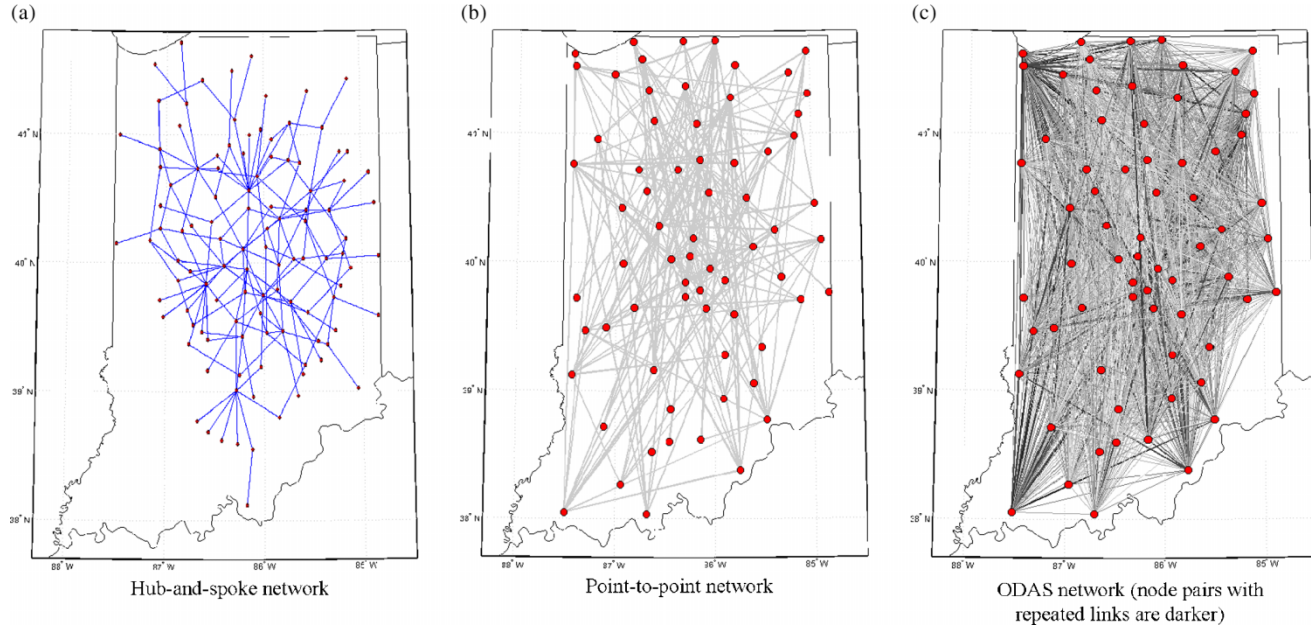


Figure 8. Representative network visualizations for the three network models: (a) hub-and-spoke network; (b) point-to-point network; and (c) ODAS network (node pairs with repeated links are darker)

grown from this initial seed. Subsequently, a pair of nodes is selected and if their multiplicative fitness (i.e. pairwise fitness defined by Eq. (3)) is greater than the average pairwise fitness of the network, a link is formed. After growing the network for a pre-specified amount of time, the final fitness of each node in the network is assessed, degree distribution calculated (see Figure 6(b)), and a visualization of the topology created (see Figure 7(b)).

The ODAS is a much more dynamic form of point-to-point service; thus, a modified evolution model is established for this case. Instead of starting with a random network (as in the hub-and-spoke or point-to-point service network), the fitness functions calculated for the point-to-point network serves to create a ‘snapshot’ network. For the ODAS network, 350 node pairs are chosen at random each time step and if their probability of linking (computed with Eq. (5)) is above a given threshold, a link is formed. The adjacency matrix of the resulting network is saved and another 350 node pairs are chosen. This process is repeated 50 times and the adjacency matrices from these runs are added together to create an aggregate network (see Figure 7(c)). Generally, the network visualizations use a binary adjacency matrix in which the links have no weight and are either connected or not. For the ODAS network, however, we use the snapshot adjacency matrix to determine the relative importance – or weight – of a link in the network. As the networks were aggregated together, node pairs which consistently linked together result in a larger value for their linkage in the adjacency matrix. These ‘heavy’ links are darker in Figure 7(b) than the links between node pairs, which seldom connected.

Simulation Results

A plot of degree distribution is useful for getting a sense of network topology type. For a random network, there is a more pronounced bell curve, denoting that there is an average number of links the network has. In the case of a hub-and-spoke network, the degree distribution follows a power-law curve, which means that there are many nodes with relatively few links and very few nodes with many links (thus the term ‘scale-free’ – there is no ‘average node’). From our simulation, we found that the hub-and-spoke network did indeed follow a power-law distribution (see Figure 6(a)), and as suspected, the point-to-point service network and ODAS were more similar to a random network (see Figure 6(b) and (c)). However, it is noteworthy to observe that the degree distribution of the ODAS network is, in a sense, an interpolation between the hub-and-spoke and scheduled point-to-point degree

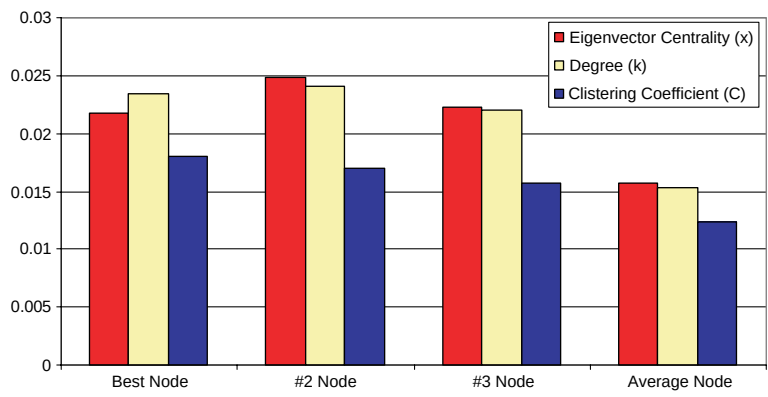


Figure 9. Relative contribution of network topology-related components to airport fitness among best performing airports in ODAS case

distributions, indicating that it may be an attractive option that combines the benefits of both types.

We are interested in determining which network topology-related components of the fitness function are predominant and how these correlate to fitness variation. A comparison of the relative impact of these components on airport fitness is offered in Figure 9. Here, the three ‘best’ airports (i.e. highest fitness at the end of the simulation) are compared against the average airport. We observe that all high performing nodes are superior to the average node by significant amounts in terms of degree, centrality, and, to a lesser extent, clustering coefficient. Further inspection indicates that eigenvector centrality and degree predominate in the distributed service networks. In the case of eigenvector centrality, there is a strong positive correlation between it and airport fitness, as shown in Figure 10. The observed relationship indicates the importance of an airport’s role in the community of airports with which it connects in driving fitness values. In other words, it ‘pays’ to be part of a good cluster in the network. This shows that the structure of the network and how the airport fits in this structure affects its success in the ODAS scenario.

The relatively simple growth models used for point-to-point and ODAS networks are effective for capturing some of the cogent dimensions of a β -level entity like an airport. However, in order to more fully (and holistically) determine the fitness of an airport, future work must include more characteristics of the airport. Fortunately, the flexibility of the fitness function formulation enables quick and intuitive integration of additional parameters.

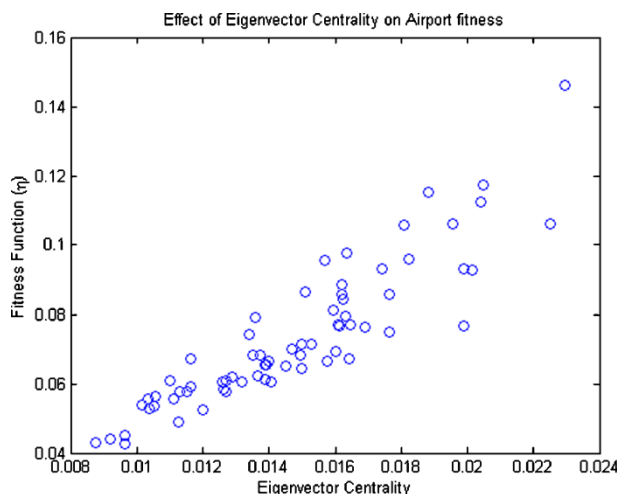


Figure 10. Correlation between nodal fitness and eigenvector centrality for each airport in the point-to-point network

Reflection and Relation to Airport Planning Approaches

The approach demonstrated in this paper may offer some avenues for improvement in airport planning over current approaches. For example, traditional airport planning seeks to forecast and deal with uncertainty often assuming the existence of current business models and operational contexts. Texts that define the airport performance and planning field, such as de Neufville and Odoni's (2003) *Airport Systems*, rely on the assumption that future performance can be assessed based on historical inference. To afford a degree of flexibility in the face of an uncertain future, aspects such as phased development must be considered in the process of dynamic strategic planning (de Neufville & Odoni, 2003). In addition, the treatment of uncertainty has begun to include options-based approaches (de Neufville, 2004). This is itself an area of great difficulty which is very fruitful for research.

However, even with these advances, the advent of ODAS and PAV may so profoundly change the environment in which airports operate that these approaches may not be adequate in scope. For instance, in the current paradigm, large airports are those which handle commercial travel and small airports are essentially considered entirely general aviation based. This polarization has led both types of airports to follow their own different types of planning practices. If more and more travel is directed to these small airports with widespread use of PAV and/or ODAS, these planning approaches may lose their validity.

The SoS approach has close ties with the socio-technical systems approach, which uses a paradigm that considers the complexities and uncertainties that result when human and technical systems are combined together. The approach for an integrated view of policy making (Walker, 2000), for example, seeks to relate these in an organized fashion. In a similar vein, Lacroix and researchers at EUROCONTROL adopt a 'large technical systems' approach, focusing on control components for decision making (Lacroix, 2001). Furthermore, Martín and Román utilize parametric analyses to benchmark airport performance beyond mere economic or operational considerations (Martín & Román, 2006). What is still needed are mechanisms for capturing and understanding the full range of technical, socioeconomic, and political dynamics which influence the behavior and environment of airports at multiple levels of hierarchy. The results presented in this paper offer one possible direction in this regard.

Conclusions

We have presented a SoS approach that provides structure to the range of problems involved in analyzing future requirements for airports. Its main tenets include a lexicon to describe the hierarchy of levels and scope of domains (including economics, operations, and policy) that comprise the problem class. The approach was explained in the context of the SoS that plays a role in the evolution of a new distributed, on-demand air transportation service paradigm. Key operational aspects of the new service were described, and a simulation was developed and instantiated with a case study of Indiana airports to show differences in the evolution of Indiana's air transport network for three different paradigms: hub-and-spoke, point-to-point, and ODAS.

We showed the role airports play in the dynamics of these networks as well as how airports with certain traits perform under the different service paradigms. Each airport has a fitness measure within each type of network, and that fitness is a measure of how the node performs and how attractive it is in the presence of different types of service providers. In particular, we found that the connectivity an airport has with other airports in the network can have a profound impact on its own relative fitness. Overall, success for an airport requires an ability to form symbiotic relationships with adjacent systems, which in turn requires an understanding of the interrelationships among these systems. The SoS approach, therefore, seems to be a promising way of designing and testing alternative airport strategies.

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